

SnapGrid: A Predictable Snap Rotation Technique with a Visible Phase Cue for Anticipatory Comfort in VR

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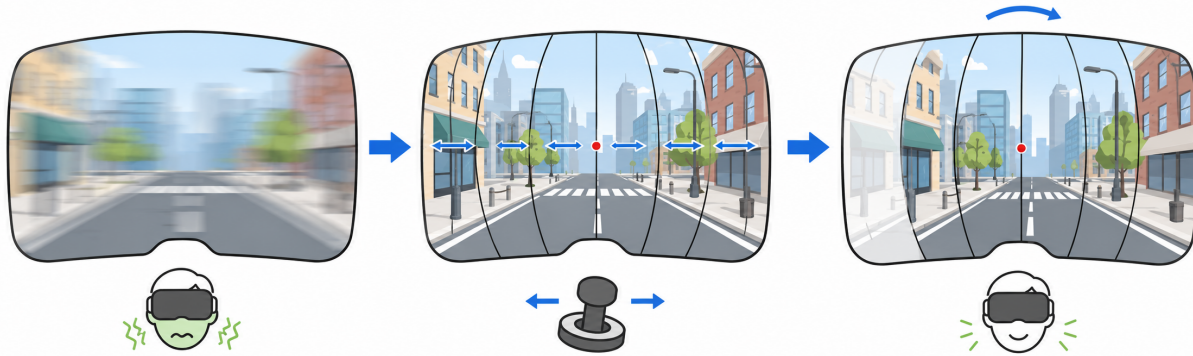


Figure 1: Concept of SnapGrid. Left: conventional continuous rotation sweeps the entire scene across the display, exposing the user to full-field optic flow that provokes cybersickness. Center: SnapGrid keeps the scene static; the rotational input instead rotates a sparse spherical grid, whose lines approach the marker fixed at the center of the view. Right: at the moment a grid line crosses the marker, the scene rotates instantaneously by one grid interval, so the user can anticipate the timing and magnitude of every snap before it occurs.

ABSTRACT

Snap rotation is a widely adopted comfort option that mitigates cybersickness by replacing continuous visual rotation with discrete viewpoint changes. Existing implementations, however, schedule these changes without informing the user in advance: while a user holds a sustained rotational input, each discrete change occurs at a moment the user cannot anticipate. We present SnapGrid, a viewpoint control technique that quantizes continuous rotational input into rhythmic snap rotations and externalizes their phase. A sparse spherical grid rotates around the camera in response to the input, and the scene rotates by one grid interval when a grid line crosses a marker fixed at the center of the view, so the timing and magnitude of every snap are perceptually available before it occurs. We describe the design rationale of SnapGrid with reference to the predictability account of cybersickness, which has so far examined anticipation cues only for passive motion, and report a pilot in-the-wild deployment in which 28 of 33 respondents rated camera rotation as more comfortable with SnapGrid enabled than disabled. We outline the controlled experiment needed to isolate the contribution of the anticipation cue.

Index Terms: Cybersickness, visually induced motion sickness, virtual reality, viewpoint control, snap rotation, anticipatory cues.

1 INTRODUCTION

Cybersickness induced by virtual viewpoint rotation remains a major obstacle to comfortable use of virtual reality. It arises when the

visual system reports a rotation that the vestibular system does not, and this mismatch is precisely a failure of the cyber and the physical to move as one: the seamless integration of the two spaces breaks down at the level of the body before any shared task can begin. Rotational optic flow provokes symptoms more reliably than translational flow, and consumer applications consequently ship comfort options that remove continuous rotation altogether, such as teleportation and snap turning. These options make VR accessible to sensitive users, but they constrain content design: tasks that require sustained camera rotation, such as following a moving target or aiming in a first-person game, map poorly onto single discrete turns triggered by button presses.

Discrete rotation itself has been studied extensively: discretizing rotational viewpoint movement can reduce sickness relative to continuous rotation [4, 5], although a recent large-scale evaluation characterized the evidence for snap turning as limited and equivocal [8]. One property of discrete rotation has received little attention in this literature: its predictability from the user's point of view. A single snap turn triggered by a button press is predictable because the user initiates it. A sequence of snaps generated while the user holds the input device deflected is not; the system schedules each jump, and the user is exposed to a series of visual discontinuities whose onsets cannot be anticipated.

Predictability is a recognized factor in motion sickness. Anticipatory cues that announce upcoming vehicle motion reduce sickness in car passengers and in driving simulators [9, 7], and unexpected vection has been shown to exacerbate cybersickness in HMD-based VR [15]. These findings, however, concern passive motion or stimulus-driven vection. To our knowledge, no interface has been designed or evaluated that provides users with an explicit visual preview of the timing and magnitude of upcoming discrete rotations under their own active control.

This paper presents SnapGrid, a technique that addresses this gap (Fig. 1). SnapGrid maps continuous rotational input onto rhythmic snap rotations and makes their phase visible: a sparse spherical grid

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rotates around the camera in response to the input, and the scene rotates by one grid interval at the moment a grid line crosses a marker fixed at the center of the view. The contributions of this paper are as follows:

- The design of SnapGrid, a viewpoint control technique that retains conventional stick and mouse operation while making the timing and magnitude of every snap rotation perceptually available in advance.
- An analysis that positions this design within the predictability account of cybersickness and identifies anticipation cues for actively controlled discrete rotation as an unexamined region of the mitigation literature.
- Descriptive questionnaire results ($N = 33$) from a pilot in-the-wild deployment of SnapGrid.

2 RELATED WORK

2.1 Discrete Viewpoint Rotation

Farmani and Teather proposed viewpoint snapping, which discretizes rotational viewpoint movement into 22.5 degree increments once rotation speed exceeds a threshold, and reported an approximately 40% reduction in SSQ scores [4]. A follow-up study extended the approach to translational movement with comparable results [5]. Ryge et al. [12] reported a preliminary investigation of discrete virtual rotation, and Sargunam and Ragan [13] compared discrete and continuous rotation for seated VR. Kelly et al. [8] evaluated field-of-view restriction and snap turning individually and in combination with 201 participants. Each technique reduced sickness in isolation, whereas the combination did not outperform either technique alone and in some cases matched the no-mitigation control. Related techniques attenuate rotational optic flow without full discretization, including rotation blurring [1] and perceptual camera control [6]. Throughout this literature, discrete jumps are treated as instantaneous events whose timing is chosen by the system; none of these techniques informs the user in advance of when the next jump will occur.

2.2 Rest Frames and Stable Visual References

The rest frame hypothesis holds that a visually stable reference reduces sickness by providing the visual system with content that is consistent with vestibular signals [11]. Duh et al. [3] showed that an independent visual background, such as an earth-fixed grid, reduces visually induced motion sickness. Cao et al. [2] found that a static, user-fixed metal net reduced sickness during HMD navigation, whereas a dynamic variant that faded with motion did not. Studies of the virtual nose report mixed results, and a recent preregistered experiment failed to replicate the original effect [16]. SnapGrid borrows the geometry of these references but inverts their role: rather than superimposing a static overlay on a moving world, SnapGrid keeps the world static between snaps and moves only a sparse overlay. The implications of this inversion are discussed in Sec. 5.

2.3 Anticipatory Cues for Upcoming Motion

Anticipatory cues have been investigated primarily in passive motion settings. Lin et al. [9] showed that motion prediction cues in a driving simulator reduced simulator sickness without degrading presence. Vibrotactile cues announcing upcoming turns reduce motion sickness in car passengers on public roads [7], and anticipatory visual, auditory, and vibrotactile displays have shown promise for capsule motion after spaceflight [10]. Consistent with these results, Teixeira et al. [15] found that unexpectedvection exacerbates cybersickness, supporting the view that visual motion is better tolerated when it can be predicted. All of this work concerns motion that the user does not control. SnapGrid transfers the anticipation cue to the active case, in which the user commands the rotation but the system schedules the discrete jumps.

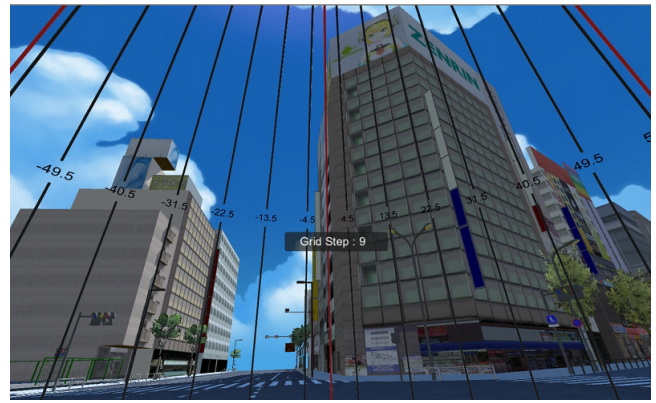


Figure 2: SnapGrid in the deployed implementation, with the grid interval θ set to 9° . A sparse spherical grid is overlaid on the scene and rotates in response to rotational input; the scene rotates by one grid interval only when a grid line crosses the red-line marker fixed at the center of the view. The labels show the angular offset of each grid line from the center of the view.

3 THE SNAPGRID TECHNIQUE

3.1 Behavior

SnapGrid replaces the camera rotation stage of a standard first-person control scheme; the user's rotational input (thumbstick deflection or mouse movement) is unchanged. The technique applies to yaw rotation only. Instead of rotating the scene camera continuously, the input rotates a sparse spherical grid centered on the camera. Grid lines are spaced at a fixed angular interval θ , which our implementation exposes in degrees as a setting that the user can adjust during play. A marker, rendered as a red line in our implementation, is fixed at the center of the display. When a rotating grid line crosses this fixed marker, the scene camera rotates instantaneously by θ in the input direction, and the grid line now aligned with the marker becomes the new reference (Fig. 2).

This design eliminates continuous full-field optic flow. Between snaps the scene is static; the only moving elements are the grid lines, which are sparse, high in contrast, and superimposed on a stationary background. The user has access to three pieces of information at all times: that the input is being registered (the grid moves), when the next snap will occur (the angular distance between the approaching grid line and the fixed marker), and the magnitude of the snap (always θ).

3.2 Implementation

The application was built with Unity 5.6.7f1 for Windows, with separate builds for the Oculus Rift DK2 (OVRPlugin SDK 0.8), the Oculus Rift CV1 (OVRPlugin SDK 1.3), and the HTC Vive (SteamVR Plugin).

The rotational input drives the grid rather than the camera. The grid rotates about the camera's vertical axis at an angular velocity proportional to the input, reaching $50^\circ/\text{s}$ at full deflection; the mapping is linear, has no dead zone, and is identical for thumbstick and mouse input. A snap of θ is executed each time a grid line reaches the marker, so the snap rate is proportional to how far the user deflects the stick, and the time the approaching line needs to reach the marker is the warning the user receives before each snap. That warning shortens as the input grows stronger. Each snap yaws the scene camera by θ in a single frame, without interpolation, and the grid line aligned with the marker becomes the new reference.

The overlay consists of meridians alone, with no parallels drawn: opaque black lines 0.01 m wide on a sphere of 3.5 m diameter centered on the camera. The marker is a red line fixed at the centre of

Table 1: Design elements of SnapGrid and the mechanisms they target.

Element	Targeted mechanism	Related evidence
Discrete scene rotation	Suppression of rotational vection via inconsistent displacement	[4, 5, 8]
Sparse moving grid in place of full-field flow	Reduction of optic flow area during input	[1, 3]
Fixed central marker	Stable fixation anchor near the fovea	[11, 2]
Grid line approaching the marker	Anticipation of snap timing and magnitude	[9, 7, 15]

the display. During play the user steps θ up or down through a fixed set of values between 1° and 90° (1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 20, 24, 30, 45, 60, 90), with 5° as the default.

3.3 Design rationale

Each element of SnapGrid corresponds to a distinct mechanism proposed in the cybersickness literature (Tab. 1).

Two properties distinguish SnapGrid from prior discrete rotation techniques. First, SnapGrid accepts sustained continuous input and converts it into a rhythmic snap sequence whose rate is proportional to input magnitude, so that tasks requiring sustained rotation remain feasible. Second, the phase of the snap cycle is externalized as a visible state, so that no jump occurs without prior perceptual warning. The grid interval θ is a single exposed parameter, which suggests applications as a per-user comfort setting or as a graduated adaptation schedule; the present paper makes no empirical claim regarding adaptation.

3.4 Potential cost of the grid overlay

The rotating grid is itself a moving visual stimulus and may induce vection or visual distraction. Whether a sparse rotating overlay presented over a static scene produces a measurable cost, and whether any such cost outweighs the benefit of the anticipation cue, is an empirical question, which we return to in Sec. 5.

4 PILOT DEPLOYMENT

The data reported here come from a public deployment that the first author conducted in 2016–2017, outside the laboratory and before the analysis in this paper was framed. It was released to a hobbyist community as a demonstration rather than designed as a hypothesis test, and we report it now because the predictability account developed in Sec. 3 identifies it as early evidence on a question the mitigation literature has left open. It is a pilot: it indicates how users respond to the technique when they choose to use it themselves, and it motivates the controlled experiment described in Sec. 5.

4.1 Study setup

We implemented SnapGrid in a VR experience application developed with Unity (Sec. 3.2) and released it publicly through Bowl-Roll, a file-distribution platform used by the Japanese creator and hobbyist community¹, with a companion blog post by the first author describing the technique². The application presented an urban scene (Fig. 2) that users explored freely with standard first-person controls, including translational movement, which SnapGrid does not modify. Users viewed the scene through their own consumer

HMDs, could switch SnapGrid on and off during camera rotation, and could adjust the grid interval during play. Users were then invited to complete an online questionnaire administered with Google Forms. Between May 2016 and May 2017, 33 responses were collected: 23 online, and 10 on paper at JPVR Meet Up #001, a meetup and exhibition for VR content developers held in January 2017, later transcribed into the same form.

The questionnaire asked for (1) self-rated susceptibility to motion sickness on a five-level scale, (2) approximate play time, (3) the comfort of camera rotation with SnapGrid disabled and (4) with SnapGrid enabled, both rated on the same eight-point ordinal scale from 1 (sickness severe enough to quit playing, or to want to quit) through 4 (neither comfortable nor uncomfortable) to 8 (extremely comfortable), (5) the most comfortable grid interval, (6) willingness to pay for the technique as a game-engine asset, and (7) free-form comments; items (2), (5), (6), and (7) were optional. It was administered in Japanese; Sec. A gives an English translation of all items together with the response counts and the free-form comments.

As a demonstration release, the deployment did not record the information a laboratory study would collect. Respondents ran the build on their own equipment. Builds were provided for the Oculus Rift DK2, the Oculus Rift CV1, and the HTC Vive, but the questionnaire did not record which headset each respondent used, and the rendering performance and physical environment are likewise unknown; the only device named in the responses is an Oculus Rift CV1, mentioned in one free-form comment. The questionnaire asked for no demographic information. Respondents set their own task and exposure time and chose when to toggle SnapGrid, leaving condition order and exposure uncontrolled and uncounterbalanced. The distribution channel had also described the intended benefit of the technique in advance. These are the conditions that unsupervised out-of-laboratory deployment imposes [14]. We therefore read the responses as evidence about acceptability, that is, whether users who tried the technique preferred it. Efficacy is addressed in the controlled experiment outlined in Sec. 5.

No research ethics committee was available to the authors at the time of the deployment, and it proceeded without ethics review, although the risks of participation were explained to users in advance. The questionnaire was anonymous and contained no explicit research-use clause; the responses are therefore reported only in anonymized and aggregated form. Their secondary use was reviewed retrospectively and approved by the Cluster Inc. Research Ethics Committee (approval number 2026-006).

4.2 Results

4.2.1 Overall comfort

Respondents covered the full range of self-rated susceptibility: 14 rated themselves more susceptible to motion sickness than average (3 of them much more), 11 as average, and 8 as less susceptible (3 of them much less). Play sessions were short: 21 respondents played for less than five minutes and 12 for five minutes or longer.

Figure 3 shows the comfort ratings. With SnapGrid disabled, no respondent chose a comfort level (5 or above); 29 of 33 reported discomfort (levels 1 to 3), including 6 at the quit level. With SnapGrid enabled, 23 of 33 chose a comfort level. Treating the levels as scores from 1 to 8, ratings were higher with SnapGrid enabled ($Mdn = 5$, $IQR = 4\text{--}6$) than disabled ($Mdn = 2$, $IQR = 2\text{--}3$), and the direction was consistent across respondents: 28 of 33 rated the enabled condition higher, 4 gave the same rating in both, and 1 rated it lower. A Wilcoxon signed-rank test rejects the hypothesis that the two labels are exchangeable (ties excluded, $n = 29$, $W = 4.0$, $z = 4.64$, $p < .001$), confirming that the asymmetry is not a chance pattern; the remaining statistics are descriptive. SnapGrid did not help everyone: 2 respondents reported quit-level sickness in both conditions.

¹<https://bowlroll.net/file/104135>

²<https://n-mattun.hatenablog.com/entry/2016/05/29/000642>

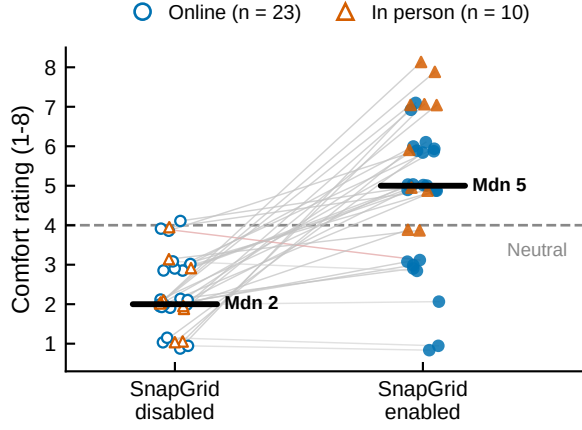


Figure 3: Ratings were higher with SnapGrid enabled than disabled for 28 of the 33 respondents (4 tied, 1 lower). Each thin line links one respondent's paired ratings; marker shape and color distinguish the online ($n = 23$) and in-person ($n = 10$) subsets. Bold bars mark the condition medians (2 and 5); the dashed line marks the neutral level (4). Condition order, exposure, and hardware were uncontrolled, and the asymmetry cannot be attributed to SnapGrid alone.

4.2.2 Exploratory breakdowns

The following breakdowns are descriptive and indicate where a controlled study should look.

The improvement was not an artifact of the in-person subset. The paper-based respondents reported larger gains (median improvement +5.0, all 10 improved) than the online respondents (median +2.0, 18 of 23 improved), as would be expected if the in-person setting inflated ratings. The pattern nonetheless holds in the anonymous online subset alone, in which 15 of 23 respondents reached a comfort level with SnapGrid enabled.

SnapGrid reached the users whom comfort options are meant to serve. Of the 6 respondents at the quit level without SnapGrid, 4 rated rotation with SnapGrid at 6 or 7, while 2 remained at the quit level. Improvements were also not confined to sickness-resistant users: respondents who rated themselves more susceptible than average improved by a median of +2.5 (12 of 14 improved), comparable to the remaining respondents (median +2.0, 16 of 19), and all 3 respondents in the most susceptible category reached a comfort level.

Most respondents would adopt the technique. Of the 25 respondents who answered the willingness-to-pay item, 23 would use SnapGrid and 2 would not; among the 23, 9 would use it only if free and 14 stated a price ceiling between 5 and 20 USD. The respondents who stated a price ceiling were also those who reported the larger gains in comfort (median improvement +4.5, against +2.0 for those who would use it only if free).

4.2.3 Preferred interval and free-form feedback

Preferred grid intervals varied widely among the respondents who gave a numeric answer (Fig. 4; $\theta = 2$ to 45 degrees, median 12 degrees); two further respondents answered qualitatively, and in opposite directions (“the narrowest value” and “wider was better”), underscoring the spread. In the free-form comments (translated from Japanese), one respondent noted that the stepwise rotation was effective for camera rotation and hard to get sick from, whereas sickness induced by lateral translation did not change with the grid displayed. Another wrote that with a narrow grid spacing the screen felt flickery and unpleasant, but with a wider spacing it was more

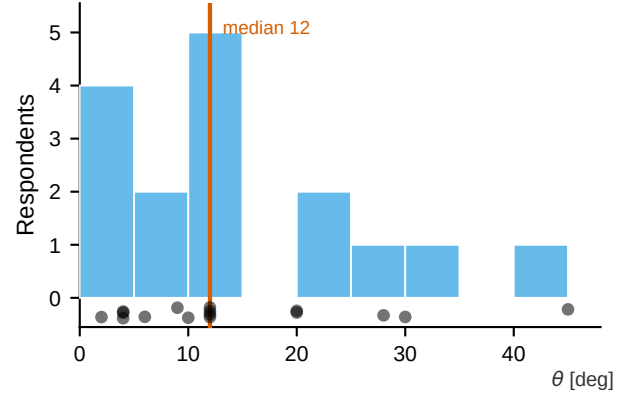


Figure 4: Preferred grid intervals spanned a wide range ($\theta = 2$ to 45 degrees, median 12 degrees), supporting exposure of θ as a per-user setting. Bars are a 5-degree histogram (left-closed bins); points below the axis show individual answers.

comfortable than expected. Several comments proposed refinements, such as keeping the rotation speed and the per-snap angle independent of the interval setting, and making the grid overlay toggleable rather than shown only while a button is held.

5 DISCUSSION

Deployment findings. Most respondents reported discomfort with unmodified camera rotation and rated rotation with SnapGrid as comfortable, and the pattern held in the anonymous online subset, among the respondents who would otherwise have stopped playing, and among those who rated themselves more susceptible than average. Users who chose to try the technique preferred rotation with it, and most of those who answered the willingness-to-pay item would adopt it. Why they preferred it is a separate question, and answering it requires the controlled experiment described below.

Design implications. Two observations bear on how the overlay should be exposed. The reported flicker at narrow grid spacings and the wide spread of preferred intervals (Fig. 4) indicate that the visual cost of the overlay depends on grid density and varies across users, which supports exposing θ as a per-user setting. The comment that SnapGrid helped rotational but not translational movement is consistent with the design targeting rotational optic flow specifically, and marks the technique’s intended scope. A further consideration follows from the input mapping (Sec. 3.2): because the snap rate is proportional to input magnitude, the warning before each snap shortens as the user pushes the stick further, and the cue may therefore behave differently at low and high angular velocities.

Toward a controlled evaluation. SnapGrid introduces four elements at once: discrete scene rotation, a sparse moving grid, a fixed central marker, and the anticipation cue that the grid’s approach provides. Their individual contributions cannot be separated in the present data. Kelly et al. [8] give reason for caution here: combining field-of-view restriction with snap turning eliminated the benefit each provided alone. We therefore plan a within-subjects experiment with five conditions: continuous rotation; snap rotation without a cue, scheduled by the system under sustained input; SnapGrid; snap rotation with an auditory cue conveying timing but neither magnitude nor direction; and conventional button-triggered snap turning, in which the user initiates each snap and its onset is predictable from the motor command alone. The fourth condition separates temporal predictability from the spatial preview the grid provides, and the fifth sets the reference point that matters

in practice, since it is the option consumer applications already ship. Because button-triggered snapping is predictable but cannot serve tasks that require sustained rotation, the experiment will measure task performance in a sustained-rotation task alongside sickness, and will manipulate angular velocity as a factor. It will use standardized measures such as the SSQ, uniform hardware, counterbalanced condition order, and controlled exposure and tasks. The same experiment will assess the cost of the overlay noted in Sec. 3.4, namely occlusion, possible vection, and reduced presence, alongside refinements such as reduced grid opacity and confinement of the grid to the visual periphery.

6 CONCLUSION

We presented SnapGrid, a viewpoint control technique that quantizes continuous rotational input into rhythmic snap rotations and externalizes their phase through a rotating spherical grid and a fixed central marker. Positioning the design against the current literature shows that anticipation cues for actively controlled discrete rotation remain unexamined. A pilot in-the-wild deployment indicates that users who tried the technique preferred rotation with it: 28 of 33 respondents rated it more comfortable, four of the six who would otherwise have stopped playing moved into the comfort range, and most respondents would adopt it. The question for the controlled experiment we have outlined is whether the anticipation cue contributes beyond discretization itself, and beyond the prediction that button-triggered snapping already affords.

SnapGrid reframes a system-scheduled discontinuity as something the user can anticipate rather than merely endure, and in doing so opens a question the anticipation literature has so far left to passive motion: how the timing, modality, and specificity of a preview should be designed when the user, not the system, initiates the movement. The grid is one point in that space, a visual and spatially explicit cue, and mapping the rest of it is a direction we hope this workshop will take up.

ACKNOWLEDGMENTS

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Supplementary Material for

“SnapGrid: A Predictable Snap Rotation Technique with a Visible Phase Cue for Anticipatory Comfort in VR”

A QUESTIONNAIRE

The questionnaire was administered in Japanese through Google Forms and was distributed together with the application. Respondents were told that answering was voluntary and anonymous. The items below are English translations; the numbering of the comfort scale is ours and was not shown to respondents. Items marked *optional* were labelled as optional in the form. Response counts over all 33 respondents are given in parentheses.

Q1. Susceptibility to motion sickness. *Are you the type who gets motion sick easily? (Your own impression is fine.)* Much more susceptible than average (3); somewhat more susceptible (11); average (11); somewhat less susceptible (5); much less susceptible (3).

Q2. Play time (optional). *Roughly how long did you play the demo application?* Less than five minutes (21); 5–10 minutes (9); 10–15 minutes (2); 15–20 minutes (1).

Q3, Q4. Comfort of camera rotation. *How uncomfortable or comfortable was camera rotation when the Grid was absent (Q3) / enabled (Q4)?* Both items used the same eight-point ordinal scale: (1) so uncomfortable that I stopped playing because of sickness, or wanted to; (2) not enough to stop playing, but I noticed sickness; (3) not sickness as such, but something felt off; (4) neither comfortable nor uncomfortable; (5) reasonably comfortable; (6) quite comfortable; (7) very comfortable; (8) amazing, almost too comfortable. Counts for levels 1 to 8 were 6, 15, 8, 4, 0, 0, 0, 0 with the Grid absent, and 2, 1, 5, 2, 9, 7, 5, 2 with the Grid enabled.

Q5. Preferred grid interval (optional, free text). *Please enter the Grid Step value you found most comfortable.* Twenty respondents entered a value. Sixteen gave a single number (2, 4, 4, 4, 4, 6, 9, 10, 12, 12, 12, 12, 20, 20, 28, 30, 45 degrees; median 12), two gave a range or an uncertain value (“30–45” and “4?”), and two answered qualitatively (“the narrowest value” and “the wider setting was better”). Figure 4 plots the sixteen single-number answers. All but one of them correspond to a selectable setting; 28° does not, and was presumably an approximate recollection.

Q6. Willingness to pay (optional). *If this camera were available on an asset store, would you buy it?* Would buy under \$20 (2); under \$10 (4); under \$5 (8); would use it if free (9); do not want it (2); no answer (8).

Q7. Free-form comments (optional). Nine respondents left comments; they are reproduced in Sec. B.

B FREE-FORM COMMENTS

All nine comments are translated below. None was omitted, and none contained information identifying the respondent.

1. When the grid was narrow the screen felt flickery, which was unpleasant, but with a wider spacing it felt more natural and more comfortable than I had expected.
2. The sickness I felt was not from camera rotation but from lateral translation, as if my head were being shaken. Displaying the Grid made no difference there. For camera rotation, the stepwise rotation was hard to get sick from and effective.
3. Thank you for sharing such an interesting experiment. One inconvenience was that the grid was not toggled on and off but had to be shown by holding the button down.

4. With the grid set to 1, rotating with the stick also made the speed increase; I think the speed should stay the same as usual. Also, with a large grid setting the angular unit by which the scene jumps becomes quite large, which felt odd; I think the angular unit should stay as it is at grid 1 even when the grid setting is changed.
5. Sickness differed when I changed the grid width according to whether I was walking or running. I would like to spend more time testing whether it also changes with the surrounding scenery.
6. I moved around to check whether rotation sickness changes with the surrounding objects, but that itself made me sick, so my later testing may have been incomplete.
7. When the Grid value is 0, rotating the camera turns the screen light blue and control is lost (CV1). When the camera faces roughly two o'clock from the initial position, performance seems to drop slightly (CV1).
8. There is a lot of room for presentation choices here; I look forward to seeing how it is applied.
9. I thought it would make people happy if this were included in many applications.